Homework 4.

Problem 2.

1. $n = kM$ 时 $k \in \mathbb{Z}$

$$e^{j\frac{2\pi n m}{M}} = e^{j\frac{2\pi k M m}{M}} = e^{j2\pi k m}$$

又因为 $e^{j\theta} = \cos\theta + j\sin\theta$, $e^{j2\pi k} = 1 \Rightarrow e^{j2\pi k m} = (e^{j2\pi k})^m = 1^m = 1$

原式 $\Rightarrow S[n] = \frac{1}{M}\sum 1 = \frac{1}{M}\cdot M = 1$

2. $n \neq kM$ 时 .

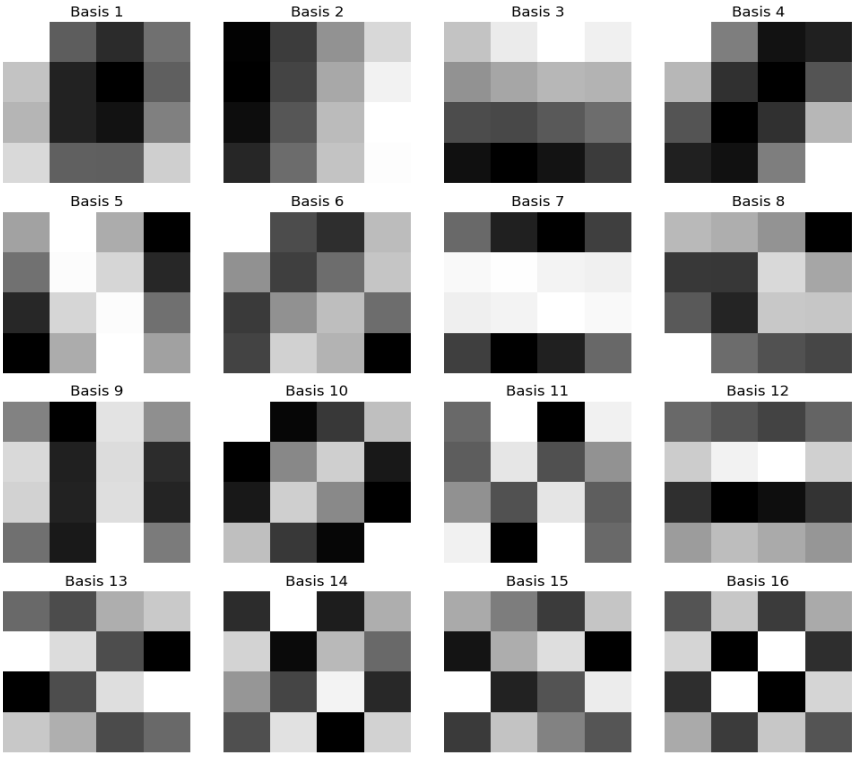
令 $r = e^{j\frac{2\pi k}{M}}$ $\quad \sum_{m=0}^{M-1} r^m = \frac{1-r^M}{1-r}$

$$r^M = \left(e^{j\frac{2\pi k}{M}}\right)^M = e^{j2\pi} = 1$$

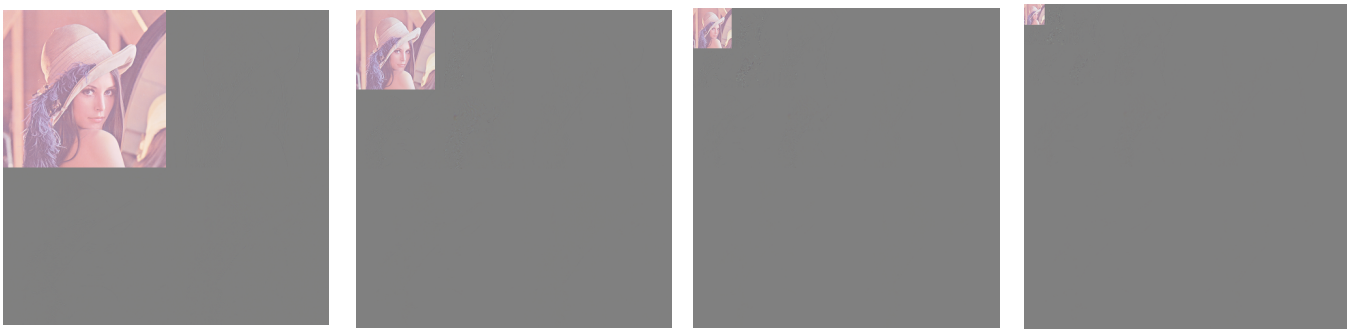
因此 $\sum_{m=0}^{M-1} r^m = 0$, 原式 $= S[n] = \frac{1}{M}\sum_{m=0}^{M-1} 0 = 0$.

综上 $S[n] = \begin{cases} 1 & n = kM,\ k \in \mathbb{Z}. \\ 0 & \text{otherwise}. \end{cases}$

Problem 1



Problem3



原图

5次小波后

从5次小波
元数据重建